Paper 18 - VOA / Moonshine Representation Routes

CQE-paper-18
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Construction Basis

Treat representation routes as upward/downward transport choices with strict receipt requirements.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-18.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-18\paper_kernel_manifest.json
- verifier: production\formal-papers\CQE-paper-18\verify_voa_moonshine_routes.py
- receipt: production\formal-papers\CQE-paper-18\voa_moonshine_routes_receipt.json (CQE Paper 18: pass)
- body: production\papers\CQE-paper-18\PAPER-BODY.md
- source: production\papers\CQE-paper-18\SOURCE.md
- formal: production\papers\CQE-paper-18\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-18\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-18\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-18.md

Paper 18 - VOA / Moonshine Representation Routes

Abstract

This paper formalizes representation routes between the finite chart VOA seed and the Monster/Moonshine boundary. The closed result is bounded and precise: the eight chart states split as `Z(q) = 2q^0 + 6q^5`, the static `Z4` route template has exactly two fixed points and six period-4 states, the Monster scalar `196883 = 47 * 59 * 71` is registered, and bounded McKay matrix bootstraps exist for the table classes `1A`, `2A`, `3A`, `5A`, and `7A`.

The paper does not prove a full Moonshine identification, does not implement `correction_via_voa`, and does not close the `O(log N)` Rule 30 extractor. It therefore treats Moonshine as a receipted route architecture with bounded evidence, not as a completed global theorem.

Claims

Claim 18.1. The finite centroid VOA seed partitions the eight chart states into two weight-0 vacua and six weight-5 excited states.

Claim 18.2. The static `Z4` representation-route template has two fixed points, zero period-2 states, and six period-4 states.

Claim 18.3. The Monster scalar used by the route is `196883`, factored in the local route table as `47 * 59 * 71`.

Claim 18.4. The bounded McKay matrix bootstrap passes for the hardcoded table classes `1A`, `2A`, `3A`, `5A`, and `7A`.

Claim 18.5. The correction-class assignment `(2,0)->2A` and `(3,1)->3A` is registered as a hypothesis, while `correction_via_voa` remains open.

Claim 18.6. The Monster-D4 lift harness provides bounded route evidence after all eight chart states activate, but reports open gaps.

Definitions

A **representation route** is a typed upward or downward transport edge between the chart seed and a larger representation boundary.

The **finite VOA seed** is the eight-state weight decomposition generated by the three-conjugate centroid labels.

The **static `Z4` template** is the four-frame route label. It is a coordinate template, not a temporal Rule 30 period claim.

A **bounded McKay bootstrap** is a finite coefficient-table and matrix receipt. It is proof-grade only at the declared bounded table size.

An **open route promotion** is any claim that requires the still-missing `correction_via_voa` evaluator, full McKay-Thompson arithmetic, or a completed Moonshine transport theorem.

Theorem 18

The CQE suite has a verified finite VOA route seed and bounded Moonshine-route bootstrap, but not a completed Rule 30/Moonshine extractor:

```
finite seed + static Z4 template + bounded McKay tables
!= full correction_via_voa route
```

Proof

The centroid VOA verifier reports `status=pass`, weight distribution `{0:2, 5:6}`, and seed partition function `Z(q) = 2q^0 + 6q^5`. This proves Claim 18.1.

The `Z4` verifier reports two fixed points, zero period-2 states, and six period-4 states. It also states that this is a static coordinate-frame template, not a temporal Rule 30 period. This proves Claim 18.2.

The VOA lookup architecture reports `MONSTER_SCALAR = 196883` and the factorization `47 * 59 * 71`. This proves Claim 18.3 as a route scalar receipt.

The McKay matrix bootstrap reports `status=pass`, honesty label `BOUNDED_EXEC`, 9-by-9 tables for all five registered classes, nested principal blocks, `3A` coefficient anchor `783`, and `2A` coefficient anchor `4372`. This proves Claim 18.4 within the bounded table scope.

The lookup harness reports that McKay coefficient parity is implemented for the bounded tables, that `correction_via_voa` is not implemented, and that the route trigger status is `WP-MOONSHINE-DEFERRED`. The direct call to `correction_via_voa` raises `NotImplementedError`. This proves Claim 18.5 and prevents the route from being promoted into a closed extractor.

The Monster-D4 lift harness reports `pass_with_open_gaps` under `BOUNDED_EXEC`: all eight chart states are enumerated, D4 lift holds after activation, and the `G2 -> F4 -> T5A` route stays within three moves. One empirical `n=3` closure subcheck remains false in that harness, so the result is bounded evidence with open gaps, not a completed Monster theorem. This proves Claim 18.6.

Together these claims prove the theorem.

Receipt

Promoted verifier:

```
production/formal-papers/CQE-paper-18/verify_voa_moonshine_routes.py
```

Receipt:

```
production/formal-papers/CQE-paper-18/voa_moonshine_routes_receipt.json
```

Closed layers:

```
finite centroid VOA sector decomposition 2q^0 + 6q^5
static Z4 route template with 2 fixed points and 6 period-4 states
Monster scalar 196883 factorization 47 * 59 * 71
bounded McKay matrix bootstrap for 1A,2A,3A,5A,7A
registered correction-class hypothesis for (2,0)->2A and (3,1)->3A
bounded Monster-D4 lift after all eight chart states activate
```

Open layers:

```
correction_via_voa implementation
full McKay-Thompson arithmetic beyond bounded tables
Rule 30 O(log N) extractor through the route
full Moonshine identification of the finite chart seed
physical representation theorem beyond the route receipts
```

Falsifiers

The paper fails if the seed partition is not `2q^0 + 6q^5`.

It fails if the `Z4` template produces period-2 states or does not split as `2 + 6`.

It fails if the bounded McKay matrix bootstrap fails.

It fails if a deferred lookup harness is presented as a completed route.

It fails if `correction_via_voa` is claimed complete.

Role in the Suite

Paper 17 supplies the code and root-shell tower that reaches the E8/Golay/Monster-facing boundary. Paper 18 supplies the finite VOA seed and bounded representation-route machinery at that boundary. Paper 19 may use the route discipline for observer-face selection only by carrying the open `correction_via_voa` and Moonshine-identification residues.

Conclusion

Paper 18 closes the finite seed and bounded bootstrap layers of the VOA/Moonshine route. It deliberately keeps the global route open until the correction evaluator, full McKay arithmetic, and Moonshine transport theorem are actually closed.